SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence **COURSE CODE:** CSA17

COURSE OUTCOME COVERED

CO3: *Analyze knowledge representation and reasoning using propositional logic, FOL, inference, resolution, chaining methods, knowledge representation to perform logical reasoning for drawing valid conclusions and decision making. (BL4)*

Assessment Tool Weightage: 50%

QUESTIONS: 30 **Time Duration: 30 Minutes**
Total Marks: 30

Annexure A
MCQ

Code of Conduct:

I J O Joanna Divine (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
Reg No: 192572003
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: JOANNA

Annexure A
MCQ – Answer Key

(Selected option circled/bolded below for each question, with a brief justification.)

Q1. Which of the following best describes a Logical Agent in AI?

Answer: (b) An agent that uses a knowledge base and logical inference to decide actions

A logical agent maintains a knowledge base and derives actions using logical inference, not randomness or pure reward-based learning.

Q2. In Propositional Logic, 'P → Q' is logically equivalent to:

Answer: (d) Both b and c

$P \rightarrow Q \equiv \neg P \vee Q$, and this is also equivalent to its contrapositive $\neg Q \rightarrow \neg P$.

Q3. Conjunctive Normal Form (CNF) requires the formula to be expressed as:

Answer: (b) A conjunction of disjunctions (clauses)

CNF is an AND of ORs — a set of clauses joined by conjunction, each clause being a disjunction of literals.

Q4. In FOL, the sentence ∀x: Student(x) → Passes(x) means:

Answer: (b) Every entity that is a student passes

The universal quantifier applies the implication to every object in the domain.

Q5. Unification in FOL is the process of finding substitutions that:

Answer: (b) Make two logical expressions syntactically identical

Unification finds a substitution of variables that makes two terms/expressions match exactly.

Q6. In the Resolution algorithm, deriving the empty clause ■ indicates:

Answer: (c) A contradiction — the original goal is proven

The empty clause means the negated goal combined with the KB is unsatisfiable, so the goal is proved true.

Q7. Forward Chaining is best described as a ________ strategy.

Answer: (b) Data-driven / bottom-up

Forward chaining starts from known facts and applies rules to derive new facts until the goal is reached.

Q8. Backward Chaining begins reasoning from:

Answer: (b) The goal and works backward to verify supporting facts

Backward chaining is goal-driven: it starts at the goal and searches for facts/rules that support it.

Q9. Which step is NOT part of converting a sentence to CNF?

Answer: (d) Distribute AND over OR

CNF conversion distributes OR over AND (not AND over OR) to produce a conjunction of disjunctions; distributing AND over OR would produce DNF instead.

Q10. Knowledge Engineering in FOL involves:

Answer: (b) Defining domain concepts, relations, and axioms formally in FOL

Knowledge engineering is the process of formally encoding a domain's objects, relations, and rules as FOL axioms.

Q11. Modus Ponens states: If P → Q is known and P is true, then:

Answer: (b) Q must be true

This is the direct definition of the Modus Ponens inference rule.

Q12. The Most General Unifier (MGU) is the substitution that:

Answer: (b) Makes two terms identical with minimal variable binding

The MGU is the least restrictive substitution that unifies two expressions.

Q13. Which inference method is more appropriate when a specific goal is known?

Answer: (c) Backward Chaining

Backward chaining works efficiently when a specific goal needs to be verified against the KB.

Q14. Propositional Logic differs from First Order Logic because it:

Answer: (b) Cannot represent objects, relations, or quantifiers — only truth values of atomic propositions

Propositional logic only deals with whole statements as true/false, with no internal structure like objects or quantifiers.

Q15. Resolution Refutation proves a goal G by:

Answer: (b) Adding ¬G to the KB and deriving a contradiction (empty clause ■)

Refutation proves G indirectly: assume ¬G, and show it leads to a contradiction with the KB.

Q16. Learning from observation in AI refers to:

Answer: (b) Improving performance by examining examples or experience from the environment

Learning from observation means an agent improves by studying examples/experience rather than only symbolic rules.

Q17. Inductive learning is best described as:

Answer: (b) Generalizing a hypothesis from specific training examples

Inductive learning builds a general rule/hypothesis from specific labeled examples.

Q18. In a Decision Tree, what does each internal node represent?

Answer: (c) A test on an attribute / feature

Internal nodes in a decision tree branch the data based on a test on one feature/attribute.

Q19. Explanation-Based Learning (EBL) differs from inductive learning because it:

Answer: (c) Uses prior domain knowledge to explain and generalize a single example

EBL uses existing domain theory to explain one training example and generalize it, needing far fewer examples than inductive learning.

Q20. In statistical learning, a Naive Bayes classifier assumes:

Answer: (c) All features are conditionally independent given the class label

The 'naive' assumption is conditional independence of features given the class.

Q21. Learning with complete data (no hidden variables) in a Bayesian network involves:

Answer: (c) Directly computing maximum likelihood estimates from observed data

With no missing/hidden variables, parameters can be estimated directly by counting/MLE from the complete data.

Q22. The Expectation-Maximization (EM) algorithm is used when:

Answer: (c) The dataset contains hidden (latent) variables that cannot be directly observed

EM iteratively estimates parameters and latent variable values when some data is unobserved.

Q23. In a Neural Network, the activation function serves to:

Answer: (c) Introduce non-linearity to enable learning of complex patterns

Activation functions add non-linearity so the network can model complex, non-linear relationships.

Q24. Which of the following correctly describes backpropagation in a neural network?

Answer: (c) The error is propagated backward from output to input layers to adjust weights

Backpropagation computes gradients of the error and propagates them backward through the layers to update weights.

Q25. A Support Vector Machine (SVM) — a type of kernel machine — classifies data by:

Answer: (c) Finding the maximum-margin hyperplane that separates classes

SVMs find the hyperplane that maximizes the margin between classes.

Q26. The kernel trick in kernel machines (SVMs) allows:

Answer: (c) Operating in a high-dimensional feature space without explicitly computing coordinates

The kernel trick computes inner products in high-dimensional space implicitly, avoiding explicit transformation.

Q27. In Reinforcement Learning, the agent aims to maximize:

Answer: (c) Cumulative reward received over time through interaction with the environment

RL agents choose actions to maximize the total (cumulative) reward over time.

Q28. The Q-Learning algorithm in Reinforcement Learning is classified as:

Answer: (c) A model-free, off-policy temporal difference learning method

Q-Learning learns action values without a model of the environment and updates off-policy using TD updates.

Q29. Overfitting in machine learning occurs when:

Answer: (c) The model fits training data too closely and generalizes poorly to new data

Overfitting means the model memorizes training data patterns/noise and performs worse on unseen data.

Q30. Which of the following best distinguishes Reinforcement Learning from Supervised Learning?

Answer: (c) RL learns via trial-and-error with reward feedback; SL learns from labeled examples

RL relies on trial-and-error with reward signals, while SL relies on labeled input-output pairs.